NBS4MED self-evaluation report

Generated: 2026-10-17

Demo version: illustrative estimates, to be reviewed when project data become available

Preliminary prototype results for the area as it is today. The methodology will be reviewed against the PROMEA framework and the NBS repository. The scores are NBS4MED prototype scores and are not an official score or certification of any external organisation or standard.

1. Assessment area

Assessment area	İzmir Green Roof Pilot
Country	Türkiye
City / municipality	İzmir
Assessment scale	Site / Project Area
Organisation	Not stated
Assessment date	2026-10-17

2. Assessment of the five criteria (provisional)

The assessment describes existing NBS integration (C1) and the enabling and implementation conditions (C2 to C5). It is provisional and separate from the suitability, impact and cost of a solution.

Completeness

13 of 14 indicators have a numeric score. 1 Unknown. 0 Not applicable.

Criterion profile

Current NBS integration (C1). How far NBS are already integrated in the area today. This criterion does not measure environmental effectiveness, such as cooling or flood reduction.

Criterion	Score	Level	Completeness and confidence
C1 NBS Integration Status	2 / 6 (33%)	Partial	3 of 3 indicators scored

Enabling and implementation conditions (C2 to C5). Policy, capacity, engagement and funding that support planning, delivery and long-term upkeep of NBS.

Criterion	Score	Level	Completeness and confidence
C2 Policy & Planning	2 / 6 (33%)	Partial	3 of 3 indicators scored
C3 Institutional & Technical Capacity	Not rated: Unknown answers	–	2 of 3 indicators scored · 1 Unknown. Available points only (not a criterion score): 2 of 4 on 2 of 3 indicators
C4 Stakeholder Engagement	2 / 4 (50%)	Partial	2 of 2 indicators scored
C5 Funding & Sustainability	1 / 6 (17%)	Insufficient	3 of 3 indicators scored

Provisional combined assessment score (secondary; not a validated measure of NBS integration)

The overall percentage is not shown because 1 indicator is Unknown (3.3). Resolve it to see the overall score.

The overall score gives every indicator equal weight, so a criterion with more indicators counts for more. Scored indicators per criterion: C1 3, C2 3, C3 2, C4 2, C5 3. Provisional project thresholds, not taken from any external standard: Insufficient below 30%, Partial 30 to 54%, Adequate 55 to 79%, Strong 80% and above. Project partners will review them.

Gaps and strengths by indicator

Priority gaps (score 0)	1.3 Link to grey infrastructure: NBS are planned separately from drainage, roads and buildings, or not linked to them at all. 2.3 Targets and monitoring: No measurable NBS targets exist. 5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS. 5.3 Co-benefits valued in decisions: Co-benefits such as cooling, health and flood protection are not counted in funding decisions.
Developing (score 1)	1.1 NBS inventory: An inventory exists but is incomplete or needs updating. 1.2 Coverage of public space: Some public space has green or blue features, but coverage is partial or uneven, or falls short of a local target where one exists. 2.1 NBS in strategic plans: NBS are mentioned in some plans, but without specific measures, locations or responsibilities. 2.2 Regulation and standards: Some rules or standards encourage NBS, but they are voluntary, partial or rarely applied. 3.1 Dedicated staff or unit: Staff handle NBS alongside other duties, or responsibility is limited to one department. 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support. 4.1 Community participation: Residents are consulted, but mostly after the main decisions have been made. 4.2 Coordination across departments and owners: Coordination happens case by case, or only between some of the actors. 5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS.
Strengths (score 2)	–
Information gaps (Unknown)	3.3 Data and monitoring tools
Not applicable	–

Priority gaps and improvement actions

Shown in a provisional display order: answers scored 0 first, then 1, then Unknown, each in indicator order. This is not a validated ranking of impact or urgency. The actions are general prompts based on your answers, not findings about the site.

#	Indicator	Type	Finding	Why it matters	Suggested action	Evidence needed
1	1.3 — Link to grey infrastructure	Gap	NBS are planned separately from drainage, roads and buildings, or not linked to them at all.	NBS planned with drainage, roads and buildings are cheaper to build, easier to maintain and more likely to work as intended.	Introduce a step in infrastructure projects to consider NBS at the design stage.	Project design documents, drainage or road plans and procedures for infrastructure projects.
2	2.3 — Targets and monitoring	Gap	No measurable NBS targets exist.	Measurable targets and regular reporting show whether NBS are progressing and allow plans to be corrected.	Define a small set of measurable NBS targets, for example area, number or condition of solutions, with a baseline year.	Target statements, monitoring indicators and progress reports.
3	5.2 — Long-term maintenance budget	Gap	No budget covers maintenance, watering or replacement of NBS.	NBS need continuing care. Without it they can fail within a few years.	Prepare an operation and maintenance budget and clarify who is responsible for funding it.	Maintenance plans, annual budget lines and financial documentation.
4	5.3 — Co-benefits valued in decisions	Gap	Co-benefits such as cooling, health and flood protection are not counted in funding decisions.	Counting co-benefits can strengthen the case for funding NBS compared with grey alternatives.	Include a short co-benefit description in funding proposals and business cases.	Funding proposals, business cases and any co-benefit studies.
5	1.1 — NBS inventory	Partial	An inventory exists but is incomplete or needs updating.	Without a record of what exists, the area cannot plan, maintain or build on its NBS.	Update and complete the inventory so that it covers the whole area and records type, location and condition.	GIS inventories, municipal records or site documentation.

#	Indicator	Type	Finding	Why it matters	Suggested action	Evidence needed
6	1.2 — Coverage of public space	Partial	Some public space has green or blue features, but coverage is partial or uneven, or falls short of a local target where one exists.	Coverage shows how much of the public space can already contribute to cooling, shade and water management.	Identify the districts with low or uneven coverage and compare coverage with a local target, or agree on one.	Land-cover or green-space maps, municipal green-area statistics and any local target.
7	2.1 — NBS in strategic plans	Partial	NBS are mentioned in some plans, but without specific measures, locations or responsibilities.	NBS that are named in plans, with measures and responsibilities, are more likely to be funded and delivered.	Add specific measures, locations and responsible bodies for NBS to the plans that mention them.	The text of the applicable plans, with the sections that mention NBS.
8	2.2 — Regulation and standards	Partial	Some rules or standards encourage NBS, but they are voluntary, partial or rarely applied.	Rules and incentives make NBS a normal part of development, not a voluntary extra.	Clarify and apply the existing rules or incentives more consistently, and track how often they are applied.	Zoning regulations, building codes, green-area standards and examples of their application.
9	3.1 — Dedicated staff or unit	Partial	Staff handle NBS alongside other duties, or responsibility is limited to one department.	Clear responsibility keeps NBS on the agenda and avoids tasks falling between departments.	Formalise the role and extend its mandate across departments.	Organisation charts, job descriptions or a decision naming the responsible unit.
10	3.2 — Technical know-how	Partial	Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.	Technical know-how decides whether NBS are built well and kept working over time.	Extend skills and guidance to further NBS types and reduce reliance on outside support.	Examples of delivered projects, guidance documents and staff or contractor qualifications.
11	4.1 — Community participation	Partial	Residents are consulted, but mostly after the main decisions have been made.	Early participation improves fit with local needs and support for upkeep.	Move participation to an earlier stage so residents can influence priorities and design.	Workshop or consultation records, and records of how input changed decisions.
12	4.2 — Coordination across departments and owners	Partial	Coordination happens case by case, or only between some of the actors.	Many NBS depend on land, assets or budgets held by different actors.	Turn case-by-case coordination into a regular mechanism, such as a standing working group.	Meeting records, agreements between actors or a working-group mandate.
13	5.1 — Capital funding	Partial	Some funding is secured or likely, but it does not cover the planned NBS.	Without capital funding, planned NBS stay on paper.	Close the funding gap for the planned NBS by securing additional sources or phasing the works.	Budget documents, grant agreements and funding commitments.
14	3.3 — Data and monitoring tools	To verify	No information is available for this indicator yet. This is a gap in information, not evidence of weak performance.	Data are needed to place NBS where they help most and to check whether they work.	Obtain or verify this information before drawing conclusions.	Dataset lists, data-sharing agreements and examples of data used in decisions.

Confidence in the answers

Of the 13 scored indicators: 0 documented evidence, 1 staff estimate, 0 informal judgement, 12 confidence not stated. Confidence is reported next to the answers. It does not change any score.

Answers in detail

ID	Indicator	Answer	Rationale	Evidence or source	Confidence
1.1	NBS inventory	1 of 2	–	Illustrative demo input	Medium: staff estimate
1.2	Coverage of public space	1 of 2	–	–	Not stated
1.3	Link to grey infrastructure	0 of 2	–	–	Not stated
2.1	NBS in strategic plans	1 of 2	–	–	Not stated
2.2	Regulation and standards	1 of 2	–	–	Not stated
2.3	Targets and monitoring	0 of 2	–	–	Not stated
3.1	Dedicated staff or unit	1 of 2	–	–	Not stated
3.2	Technical know-how	1 of 2	–	–	Not stated
3.3	Data and monitoring tools	Unknown	–	–	Not stated
4.1	Community participation	1 of 2	–	–	Not stated

ID	Indicator	Answer	Rationale	Evidence or source	Confidence
4.2	Coordination across departments and owners	1 of 2	–	–	Not stated
5.1	Capital funding	1 of 2	–	–	Not stated
5.2	Long-term maintenance budget	0 of 2	–	–	Not stated
5.3	Co-benefits valued in decisions	0 of 2	–	–	Not stated

Is there a local target for NBS or for green and blue space coverage? No. Is the current coverage known (mapped or measured)? No.

Site context (not scored)

Urban Density	High
Public Land Available for NBS	Limited
Local Climate Risks	Heatwaves; Flooding

How the assessment scores are calculated (provisional)

- Each indicator is scored 0, 1 or 2 using the provisional project rubric shown in the assessment.
- Unknown is missing information and is never counted as 0. Not applicable (with a justification) is also left out. Both are reported separately.
- A criterion percentage is the sum of its scored indicators divided by 2 times the number of scored indicators. It is shown only when the criterion has no Unknown answer. A criterion whose indicators are all Not applicable is shown as Not applicable.
- The overall percentage uses the same rule across all indicators and is shown only when no indicator is Unknown. With every indicator scored, it is the total divided by 28. Every indicator has equal weight, so criteria with more indicators weigh more.
- Levels use provisional project thresholds: Insufficient below 30%, Partial 30 to 54%, Adequate 55 to 79%, Strong 80% and above.
- The scores are not multiplied by confidence. Confidence is reported separately, indicator by indicator.
- C1 describes current NBS integration only. It is not a measure of environmental effectiveness.

3. Applicability screening

Screening only, not engineering approval. These checks do not change the assessment score or the match score. Answers about planting space, soil and water are shared by the planting-based options.

Option	Category (provisional)	Screening status	Answer
Green roof	Green infrastructure	Requires verification	Is there a roof in the area that could potentially host a green roof? Yes | What is the structural suitability of that roof? Assessment pending or Unknown

4. Candidate solutions and reasons

Options are listed in the order of a provisional rule-based match to your selected risks and site context. This is not a validated ranking. Applicability statuses come from optional checks and are not engineering approval. The match is a provisional rule-based description of how well an option addresses the selected risks and the land context. It is not a probability of success, a feasibility assessment or a predicted climate impact.

#	Option	Category (provisional)	Rule-based match score	Screening status	Why it matches the selected risks	To verify	Implementation conditions
1	Green roof	Green infrastructure	67 / 100	Requires verification	Heatwaves (moderately); Flooding (moderately). Uses existing roof space, not public land. Roof availability and structure still need checking.	Structural suitability of the roof requires assessment (assessment pending or Unknown).	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
2	Green corridor	Green infrastructure	67 / 100	Not yet checked	Heatwaves (strongly); Flooding (weakly). Land need fits the available public land.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
3	Urban forest (Miyawaki method)	Green infrastructure · Ecosystem restoration	52 / 100	Not yet checked	Heatwaves (strongly); Flooding (weakly). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
4	Rain garden and bioswale	Blue-green water management	52 / 100	Not yet checked	Heatwaves (weakly); Flooding (strongly). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
5	Permeable paving	Blue-green water management · Hybrid/integrated solutions	50 / 100	Not yet checked	Flooding (strongly). Not a target of this option: Heatwaves. Land need fits the available public land.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.

#	Option	Category (provisional)	Rule-based match score	Screening status	Why it matches the selected risks	To verify	Implementation conditions
6	Smart eco-park	Hybrid/integrated solutions · Green infrastructure	43 / 100	Not yet checked	Heatwaves (strongly); Flooding (moderately). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
7	Pocket park	Green infrastructure	35 / 100	Not yet checked	Heatwaves (moderately); Flooding (weakly). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
8	Rainwater retention basin	Blue-green water management · Hybrid/integrated solutions	27 / 100	Not yet checked	Heatwaves (weakly); Flooding (strongly). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.
9	Vegetated dune and shoreline	Ecosystem restoration	2 / 100	Not yet checked	Flooding (weakly). Not a target of this option: Heatwaves. Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.1 Capital funding: Some funding is secured or likely, but it does not cover the planned NBS. | 5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS. | 3.2 Technical know-how: Some experience or guidance exists, but it is limited to certain NBS types or relies on outside support.

How the rule-based match score is calculated (provisional)

- Each option has a weight from 0 (not a target) to 3 (strong) for heatwaves, flooding and drought. These are the earlier demo weights, kept as provisional rules pending methodology review. They have not been validated by PROMEA.
- Matching rules for Coastal Storms and Land Degradation have not been defined. When either is selected, or when Not Sure is selected, no match score or ranking is produced.
- Score = sum of the weights for your selected risks, divided by 3 times the number of selected risks, times 100.
- Land context: 15 points are subtracted for each level by which the option's land need exceeds the public land available, and 10 more points when urban density is high and the land need is high. The score is limited to 0 to 100.
- Urban density and public land availability are qualitative answers, not numerical thresholds. If public land availability is Unknown, no match score is produced. Institutional gaps (funding, maintenance, technical know-how) are not used to change the score. They are shown as implementation conditions.

- Climate type is not yet used in matching. It will be considered once verified location-specific information and decision rules are available.
- Public land availability is not used as evidence that a roof is available. Roofs are checked separately in the applicability checks.
- Screening: an option is Not applicable only when a basic condition is answered No (for example no roof, no coast, no runoff source). Unknown, or a No on a condition that can be assessed or designed (for example structural suitability), gives Requires verification.

Option	Heatwaves	Flooding	Drought
Urban forest (Miyawaki method)	3	1	1
Green roof	2	2	1
Green corridor	3	1	0
Rain garden and bioswale	1	3	1
Permeable paving	0	3	1
Pocket park	2	1	0
Smart eco-park	3	2	2
Rainwater retention basin	1	3	3
Vegetated dune and shoreline	0	1	0

Working alignment with the ten PROMEA considerations

Working framework used by the project team. This is not a formal PROMEA validation. Items marked Pending are not yet operational in this prototype.

Consideration	Status	Detail
Climate Risks	Partially supported	Selection of five risks (multiple allowed, plus Not Sure) is implemented. Provisional matching rules exist only for heatwaves, flooding and drought.
Climate Type	Pending further development	Not asked and not inferred. Needs verified location-specific information and decision rules.
Urban Context	Partially supported	Urban density and public land availability are asked as qualitative answers and used in the provisional matching.
NBS Categories	Partially supported	Provisional category labels on each option. They do not affect matching.
Cost	Partially supported	Indicative demo cost ranges for five of the nine options, not local prices.
Stakeholders	Partially supported	C4 Stakeholder Engagement (4.1, 4.2) on the provisional 0 to 2 rubric. Not used in matching.
Regulations	Partially supported	C2 Policy & Planning (2.1 to 2.3) on the provisional 0 to 2 rubric. Not used in matching.
Technical Expertise	Partially supported	C3 Institutional & Technical Capacity (3.1 to 3.3). Technical know-how (3.2) is also shown as an implementation condition.
Maintenance	Partially supported	Indicator 5.2, shown as an implementation condition, and an annual maintenance cost range for supported options.
Evaluation	Pending further development	Indicators 2.3 and 3.3 are assessed, but there is no monitoring or evaluation module for NBS performance.

5. Impact and cost exploration (illustrative)

Green roof

Demo version: illustrative estimates, to be reviewed when project data become available

From your assessment: Relevant to: **Heatwaves; Flooding**. **Requires verification**

- Structural suitability of the roof requires assessment (assessment pending or Unknown).

Illustrative outputs

Results change with your inputs. Each group says what kind of figure it is: arithmetic, a literature-based scenario, or a demo assumption.

Roof structural suitability requires verification. These figures assume the roof can carry the system, which has not been checked.

Quantities calculated from your inputs

Indicator	Result	Unit	Basis
Rain falling on the roof in this event Context	24	m^3	**Arithmetic on your inputs**
Substrate volume to buy Design quantity	64	m^3	**Arithmetic on your inputs**

Arithmetic on your inputs and on demo design assumptions. These are quantities, not impact forecasts.

Literature-based scenario

Indicator	Result	Unit	Basis
Rainwater retained in this single event (scenario) Single-event retention Published range: 26.8 to 61.8% of the event rainfall.	6.43 – 14.8	m^3 (one event)	**Published reference**

Literature-based event retention is an illustrative scenario, not a validated local forecast. It applies published results from other places to your roof.

Demo cost assumptions

Funding and maintenance conditions from your assessment

- **5.1 Capital funding**: Score 1: Some funding is secured or likely, but it does not cover the planned NBS.
- **5.2 Long-term maintenance budget**: Score 0: No budget covers maintenance, watering or replacement of NBS.

Currency EUR. No price year: demo prices were not taken from a dated price list. Not a local cost.

Construction cost (one-off)

Item	Quantity	Unit price (low / base / high)	Subtotal (low to high) (EUR)	base (EUR)
Growing substrate **Demo assumption**	64 m^3	60 / 90 / 140 EUR/m^3	3,840 – 8,960	5,760
Drainage and protection layers **Demo assumption**	800 m^2	5 / 8 / 14 EUR/m^2	4,000 – 11,200	6,400
Vegetation **Demo assumption**	800 m^2	6 / 12 / 25 EUR/m^2	4,800 – 20,000	9,600
Installation labour **Demo assumption**	800 m^2	4 / 8 / 15 EUR/m^2	3,200 – 12,000	6,400
Root-resistant waterproofing **Demo assumption**	800 m^2	8 / 14 / 25 EUR/m^2	6,400 – 20,000	11,200
Structural check and design **Demo assumption**	1 lump sum	1,000 / 2,500 / 6,000 EUR (lump sum)	1,000 – 6,000	2,500

Item	Quantity	Unit price (low / base / high)	Subtotal (low to high) (EUR)	base (EUR)
Total			**23,240 – 78,160**	**41,860**

Annual maintenance (per year)

Item	Quantity	Unit price (low / base / high)	Subtotal (low to high) (EUR)	base (EUR)
Maintenance Demo assumption	800 m^2	1 / 2.5 / 5 EUR/m^2/year	800 – 4,000	2,000
Total			**800 – 4,000**	**2,000**

Illustrative scenario range: Low and high are alternative demo assumptions. This is not a statistical confidence interval.

Includes: the items listed above only. Excludes land acquisition, VAT unless stated, utility relocation, contingency, financing and replacement beyond annual maintenance.

Lump-sum items do not change with area, and unit prices do not fall with scale. Real costs rarely grow in a straight line with area.

Reference-case cost scaling (Hamburg, extensive roofs)

Cost per m^2 of total roof area in 7 extensive roofs of one study, multiplied by your area. This is a source example, not a local cost and not a forecast.

	EUR/m^2	EUR
lowest case	29	23,200
median of cases	44	35,200
highest case	73	58,400

Currency EUR; prices of 2017 (May 2017 construction index); VAT included; creation and establishment care; root-resistant waterproofing excluded; area basis: total roof area.

▶ **Observed cases (7)**

The author also gives a descriptive band of 40 to 45 EUR/m^2 for most roofs (printed p.14). It is not an observed minimum and maximum. One intensive-roof case (58 EUR/m^2, 591 m^2) is not mixed with the extensive cases.

Reference costs from another place and year are not current local prices.

Main assumptions and limitations

- All figures are illustrative and for early planning only. They are not a quote, a design or a forecast for your site.
- Demo values are placeholders until project partners review them.
- Roof structural suitability requires verification. These figures assume the roof can carry the system, which has not been checked.
- Literature-based event retention is an illustrative scenario, not a validated local forecast. It applies published results from other places to your roof.
- Reference costs from another place and year are not current local prices.
- Published evidence from other places is listed under Methodology and sources. It is not a forecast for your site.

Not calculated, and why

- **Annual rainwater retention.** Event retention is not annual retention. Annual values need a rainfall series and the moisture state of the roof.

- **Reduction of peak discharge.** It needs outlet design and hydrologic modelling, which this demo does not include.
- **Change in air temperature at your site.** No intervention-specific coefficient is available. A published contrast is a measured difference in one study, not a prediction.
- **Change in surface temperature.** No surface-temperature record has been collected.

Information needed to improve the estimate

- **Site checks proposed for review:** Confirm structural capacity, waterproofing, drainage and vegetation.
- **Cost information:** Local item prices, quantities and specification
- **Impact information:** Roof system, rainfall series and substrate moisture
- Validated definitions, source review and ranking rules from project partners; the NBS repository and the PROMEA decision-making framework.

▼ Methodology and sources

Technical details: formulas, parameter keys, rule codes, published evidence and sources.

Formulas used

Indicator	Formula
Rain falling on the roof in this event	`V_rain = A × h / 1000` F-N02-RAIN
Rainwater retained in this single event (scenario)	`V_ret = V_rain × r` F-N02-RET
Substrate volume to buy	`V_sub = A × t` F-N02-SUB

Published reference evidence (not a forecast)

Published examples from other places. They are shown to give context. They are not predictions for your site and are never added together.

2 independent studies behind 11 records. Not usable as a local forecast (usable_for_local_forecast = false).

Retention in single rainfall events

Metric	Value	Statistic	Place and year	Study	Limits	Source
Mean per-event rainfall retention for events >10 mm G01	26.8 – 61.8 percent	Range of configuration means	UK experimental roofs, 2015	S06:uk-experimental-roofs	Range of bed-specific means; not annual retention or prediction interval	S06 abstract; DOI 10.1016/j.ecoleng.2015.09.076

Metric	Value	Statistic	Place and year	Study	Limits	Source
Typical retention for events <10 mm G02	80 percent; greater than	Qualitative threshold	UK experimental roofs, 2015	S06:uk-experimental-roofs	Typical >80%, not a guaranteed lower bound	S06 abstract

Construction cost examples

Metric	Value	Statistic	Place and year	Study	Limits	Source
Extensive roof: author descriptive typical band E02	40 – 45 EUR/m² total roof	Author's descriptive band (not observed min and max)	Hamburg, Germany, 2017	S04:hamburg-green-roofs	Printed p.14: descriptive band for most roofs, not observed min/max; see individual cases	S04 printed pp.10–15, Fig.3

Inputs and assumptions

Your inputs

Roof area to green	800 m²	Your value
Design rainfall event depth	30 mm	Your value
Substrate depth	0.08 m	Demo default
Include new root-resistant waterproofing	✓	Demo default

Assumptions used (parameters)

Parameter	Value	Class	Valid range	Partner validation
Growing substrate NBS02.substrate_price	60 / 90 / 140 EUR/m³	Demo assumption	50 – 5000 area_m2	Not yet validated
Drainage, filter and protection layers NBS02.drainage_layer	5 / 8 / 14 EUR/m²	Demo assumption	50 – 5000 area_m2	Not yet validated
Vegetation (mats or plugs) NBS02.vegetation	6 / 12 / 25 EUR/m²	Demo assumption	50 – 5000 area_m2	Not yet validated
Installation labour NBS02.install_labour	4 / 8 / 15 EUR/m²	Demo assumption	50 – 5000 area_m2	Not yet validated
Root-resistant waterproofing NBS02.waterproofing	8 / 14 / 25 EUR/m²	Demo assumption	50 – 5000 area_m2	Not yet validated
Structural check and design NBS02.structural_check_lump	1,000 / 2,500 / 6,000 EUR (lump sum)	Demo assumption	50 – 5000 area_m2	Not yet validated
Annual maintenance (extensive roof) NBS02.maintenance_annual	1 / 2.5 / 5 EUR/m²/year	Demo assumption	50 – 5000 area_m2	Not yet validated

Parameter	Value	Class	Valid range	Partner validation
Mean per-event retention, events > 10 mm NBS02.event_retention_gt10mm	G01	Published reference	Experimental roofs, UK, 80 mm extensive build-up; events > 10 mm. Upper event depth not extracted.	Not yet validated
Extensive-roof construction cost cases (7) NBS02.reference_cost_extensive	H1, H2, H4, H6, H7, H8, H9	Published reference	Case roof areas 196 to 1,944 m^2; Hamburg; VAT included; root-resistant waterproofing excluded	Not yet validated

Rules applied

- R01: site information is insufficient, so site verification is required.
- R02: structural capacity is unknown, so technical feasibility is not claimed.
- R06: source context or price year differ from your site, so published values are source examples only.
- R08 and R09: different outcomes and statistics are never pooled; records of one study count as one study.
- R10: a missing value is reported as unavailable, never as zero.

Evidence sources

- S02: Climate-ADAPT: Water sensitive urban and building design (EEA / European Commission). Costs and benefits; original studies linked on page https://climate-adapt.eea.europa.eu/en/metadata/adaptation-options/water-sensitive-urban-and-building-design
- S03: Climate-ADAPT: Climate proofing of buildings against excessive heat and cold (EEA / European Commission). Costs and benefits; Hamburg roof example https://climate-adapt.eea.europa.eu/en/metadata/adaptation-options/climate-proofing-of-buildings-against-excessive-heat
- S04: Hamburg's Green Roofs: An Economic Evaluation (BUE / HafenCity University, 2017). Original report; printed pp.10–15, Fig.3 https://www.hamburg.com/resource/blob/1057968/4f3ba7ede2a8a5e1038f73e5f24a75b9/d-economic-evaluation-data.pdf
- S06: The influence of substrate and vegetation configuration on green roof hydrological performance (Stovin, Poë, De-Ville & Berretta, 2015). Author institutional record: abstract reviewed; full tables not extracted https://eprints.whiterose.ac.uk/id/eprint/93038/

Demo version: illustrative estimates, to be reviewed when project data become available